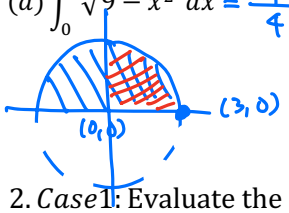
$y=\sqrt{9-x^2} \Rightarrow y^2=9-x^2 \Rightarrow x^2+y^2=9$

MAT1575, Classwork9, Fall2026

3.3 Trigonometric Substitution (p.250-259)

1. Evaluate each of the following integrals:

(a) $\int_0^3 \sqrt{9-x^2}\,dx = \frac{9}{4}\pi$



(b) $\int \frac{x}{\sqrt{9-x^2}}dx$?

$1-\sin^2\theta=\cos^2\theta$

let $x=3\sin\theta$ (why?)

$\sqrt{9-x^2}=\sqrt{9-9\sin^2\theta}=\sqrt{9\cos^2\theta}=3|\cos\theta|$?

(c) $\int x\sqrt{9-x^2}\,dx$?

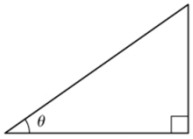
let $x=3\sin\theta$?

2. *Case1*: Evaluate the given indefinite integral $\int \sqrt{9-x^2}\,dx$ *Hint*: *let* $x = 3\sin\theta$

**Check the angle based on x:*

Under this range of θ, we have

**How to put x back to the result*:



3. *Problem-Solving Strategy*: **Integrating Expressions Involving $\sqrt{a^2-x^2}$**

(1) Make the substitution $x =$ and $dx =$ which yields $\sqrt{a^2-x^2} =$

(2) Simplify the expression.

(3) Evaluate the integral using techniques from the section on trigonometric integrals.

(4) Use the reference triangle to rewrite the result in terms of x: $\theta =$

4. Evaluate $\int \frac{\sqrt{4-x^2}}{x}dx$

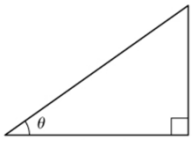
5. Evaluate $\int x^3\sqrt{1-x^2}\,dx$ two ways: (1)using $u = 1-x^2$, (2)using a trigonometric substitution.

6. $Case2$: Evaluate the given indefinite integral $\int \frac{dx}{\sqrt{1+x^2}}$ $Hint: let\ x = \tan\theta$

Check the angle based on x:

Under this range of θ, we have

How to put x back to the result:



7. *Problem-Solving Strategy*: **Integrating Expressions Involving $\sqrt{a^2+x^2}$**

(1) Make the substitution $x =$ and $dx =$ which yields $\sqrt{a^2+x^2} =$

(2) Simplify the expression.

(3) Evaluate the integral using techniques from the section on trigonometric integrals.

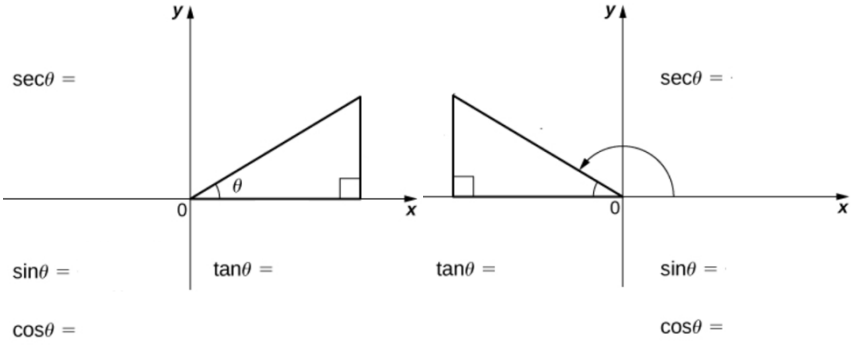
(4) Use the reference triangle to rewrite the result in terms of x: $\theta =$

8. $Case3$: Evaluate the given indefinite integral $\int \sqrt{x^2-9}\,dx$ $Hint: let\ x = \sec\theta$

Check the angle based on x:

Under this range of θ, we have

How to put x back to the result:



9. *Problem-Solving Strategy*: **Integrating Expressions Involving $\sqrt{x^2-a^2}$**

(1) Make the substitution $x =$ and $dx =$ which yields $\sqrt{x^2-a^2} =$

(2) Simplify the expression.

(3) Evaluate the integral using techniques from the section on trigonometric integrals.

(4) Use the reference triangle to rewrite the result in terms of x: $\theta =$

10. Find the area of the region between the graph of $f(x) = \sqrt{x^2-9}$ and x-axis over the interval $[3, 5]$.